ITCS544 Term Project — Proposal Pitch

Working title: Satellite Fire Detection Warehouse for Monitoring Agricultural Land Conversion

Domain: Agriculture / land-use change (spatio-temporal, IoT-style sensor feeds)

The pitch in one paragraph

Agricultural expansion into forest and savanna is driven by burning. Satellites detect those fires several times a day, globally, and NASA publishes every detection as an open CSV going back to 2000. We build a multi-tier data warehouse over roughly 20–40 million of these detections for a single region, conform them against protected-area boundaries and vegetation-index time series, and then mine the result: clustering individual detections into discrete fire events, and detecting anomalous burning inside legally protected areas. Every requirement in the assignment maps onto a concrete piece of this system, and the data is genuinely raw — multi-satellite, inconsistent schemas, duplicate detections, and records that get retroactively replaced months after publication.

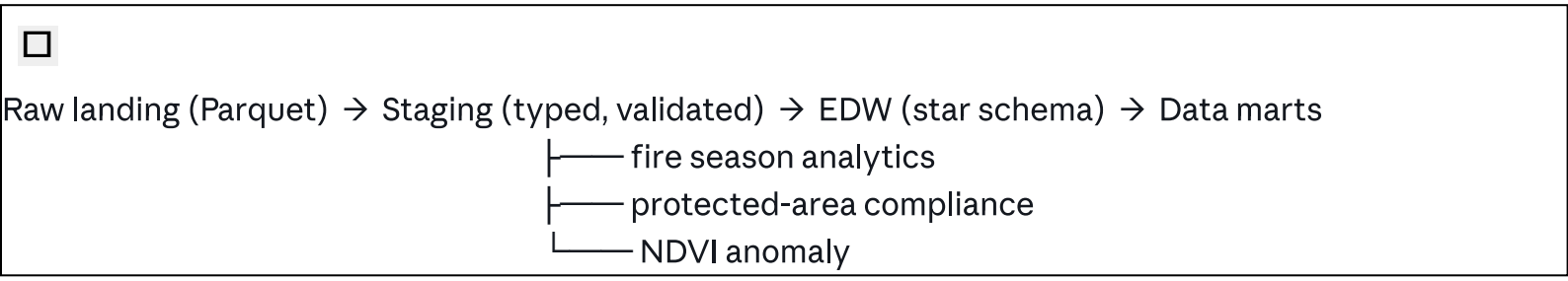
Why this domain

The data is already row-shaped. No rasterisation, no manufactured row counts. Each satellite detection is one record with a timestamp and coordinates.
The messiness is real, not contrived. MODIS and VIIRS use different column layouts. The same fire is detected by up to five satellites on overlapping overpasses. Near-real-time records are replaced by science-quality records after roughly a five-month lag. That is three independent data quality problems handed to us for free.
It overlaps with our GIS coursework, so we start with the spatial skills already in hand.
The results mean something. "Unauthorised burning inside a protected area" is an actual finding, not a toy classification score.

Research questions

Can we cluster raw detections into coherent fire events, and how does clustering quality vary across biome and season?
What fraction of detected burning falls inside protected areas, and has that fraction changed over time?
Can we predict anomalous burning at municipality-month granularity from historical patterns and pre-fire vegetation state?

Architecture



Orchestration with Airflow, transformations in PySpark, warehouse in Postgres (PostGIS) or DuckDB, everything containerised with Docker Compose.

Fact table grain: one row per satellite detection. **Measures:** brightness, fire radiative power, scan, track, confidence. **Dimensions:** date, acquisition time, satellite/instrument, location (H3 cell → municipality → state → biome), protected area (SCD Type 2), land cover (SCD Type 2), junk dimension for day/night + processing version. **Degenerate dimension:** composite natural key built from satellite + timestamp + coordinates, since FIRMS ships no unique detection ID.

The data files

Note: column lists below are from documentation, not from files we have opened yet. First task for whoever takes data acquisition is to download one file per source and confirm the actual headers against the official manuals. Expect small differences.

1. FIRMS active fire archive — our fact source

Downloaded from the NASA FIRMS archive download page (free, Earthdata login). Separate file per instrument per region per date range.

File pattern	Instrument	Coverage starts	Resolution
`fire_archive_M-C61_*.csv`	MODIS Terra + Aqua	1 Nov 2000	~1 km
`fire_archive_SV-C2_*.csv`	VIIRS S-NPP	20 Jan 2012	375 m
`fire_archive_J1V-C2_*.csv`	VIIRS NOAA-20	1 Apr 2018	375 m
`fire_archive_J2V-C2_*.csv`	VIIRS NOAA-21	17 Jan 2024	375 m

Roughly: `latitude`, `longitude`, `acq_date`, `acq_time`, `satellite`, `instrument`, `confidence`, `version`, `frp` (fire radiative power), `daynight`, `type`, `scan`, `track`, plus a brightness temperature pair whose column names **differ between MODIS and VIIRS** (`brightness` / `bright_t31` versus `bright_ti4` / `bright_ti5`). Reconciling these into one conformed staging table is a real chunk of our ETL story.

`confidence` is also encoded differently — a 0–100 integer on MODIS, a categorical low/nominal/high on VIIRS. Needs an explicit harmonisation rule.

Also grab: the yearly summary by country CSV from the same page. Small, and it tells us exact record counts per year per country so the proposal can state a real number.

2. WDPA — our SCD Type 2 dimension

`WDPA_Aug2026_Public_csv` — attribute table, no geometry.

Key fields: `WDPAID`, `WDPA_PID`, `PA_DEF`, `NAME`, `DESIG_ENG`, `DESIG_TYPE`, `IUCN_CAT`, `MARINE`, `REP_AREA`, `STATUS`, `STATUS_YR`, `GOV_TYPE`, `OWN_TYPE`, `MANG_AUTH`, `VERIF`, `PARENT_ISO3`, `ISO3`.

Critical: `WDPAID` identifies a protected area, `WDPA_PID` identifies a parcel or zone within one. Multi-part areas produce several rows sharing one `WDPAID`. Natural key must be `WDPA_PID`.

We also need the file geodatabase version (~1.1 GB, global, geometry included). The CSV alone cannot support the point-in-polygon join that the whole protected-area analysis depends on. Take both: geometry from the gdb, cheap monthly diffing from the CSV.

Historical versions for the SCD2 change history come from Earth Engine assets (`WCMC/WDPA/YYYYMM/polygons`, monthly from July 2017), because protectedplanet.net only serves the current release. Fallback if that proves awkward: build a weaker Type 2 from `STATUS_YR` alone.

Licensing: WDPA cannot be redistributed without written permission from UNEP-WCMC. It goes in `.gitignore`; the repo ships a download script and the citation instead.

3. MODIS NDVI (MOD13Q1) — vegetation state

Extracted via AppEEARS as CSV, 16-day composites at 250 m from 2000. Columns approximately `Latitude`, `Longitude`, `Date`, `MOD13Q1_061__250m_16_days_NDVI`, plus VI quality and pixel reliability layers. AppEEARS applies the scale factor in CSV output (the raw rasters do not).

This file is load-bearing for one specific requirement. The FIRMS records are largely complete, so they give us almost nothing to impute. Cloud-obscured NDVI composites are where our missing-value imputation story lives — temporal interpolation or Savitzky-Golay smoothing over gappy time series.

4. Administrative boundaries

Municipality and state polygons for the chosen region (IBGE for Brazil, or equivalent). Shapefile, not CSV. Feeds the location hierarchy and gives us a second SCD Type 2 candidate, since municipality boundaries get redrawn over time.

How this maps to the grading requirements

Requirement	Where it lives
5M+ raw records	FIRMS multi-satellite archive, regional scope
Multi-tier architecture	Raw → staging → EDW → 3 marts
Star schema + SCD2 + degenerate dim	Detection fact; WDPA and land cover as SCD2; composite detection key
Schema validation	MODIS vs VIIRS column reconciliation
Deduplication	Cross-satellite overlap **and** NRT superseded by science-quality
Missing value imputation	NDVI cloud gaps

Requirement	Where it lives
Anomaly detection in ETL	Gas flares and volcanoes (known false positives), impossible coordinates
OLAP	Space hierarchy + dual calendar (standard year and Aug–Jul fire year)
Mining task 1	HDBSCAN space-time clustering into fire events
Mining task 2	Isolation Forest on municipality-month aggregates + protected-area violations
Metrics	Silhouette, ROC-AUC, PR curves, throughput at 1M/5M/20M rows

Division of work

Three workstreams, one owner each. They overlap heavily after the progress report — nobody owns a silo through to the end, and everyone writes part of the final report.

Workstream	Scope	Owns these deliverables
A — Ingestion & ETL	FIRMS / WDPA / NDVI extraction, schema reconciliation across MODIS and VIIRS, deduplication rules, NRT-versus-science-quality supersession, imputation, Airflow DAG	ETL documentation, before/after quality metrics
B — Warehouse & OLAP	Star schema, SCD Type 2 implementation, surrogate key generation, partitioning and spatial indexing, query tuning	ERD, bus matrix, 5 benchmarked OLAP queries
C — Mining & evaluation	Feature engineering off the marts, HDBSCAN clustering, Isolation Forest, metrics, plots	EDA, baseline models, evaluation section

Cross-cutting and shared: Docker Compose setup, README, the architecture diagram, and rehearsing the live demo.

Timeline

Date	Deliverable	Weight	Internal target
1 Sep	Proposal	15%	Region locked, exact row count from FIRMS country summary, architecture diagram drawn
13 Oct	Progress report + live demo	25%	Warehouse loaded end to end, DAG running unattended, 5 OLAP queries benchmarked, one baseline model

Date	Deliverable	Weight	Internal target
17 Nov	Presentation	15%	Demo rehearsed twice against a cold start
30 Nov	Final report + repo	45%	Docker Compose verified on a clean machine, README replication steps tested by someone who did not write them

Weekly milestones to 13 October:

Wk 1 (1–7 Sep) — Region locked. All four sources downloaded, real headers verified against docs.
Wk 2 (8–14 Sep) — Raw landing zone populated. Staging schema drafted. Dedup rules specified.
Wk 3 (15–21 Sep) — Star schema built. Surrogate keys and SCD2 logic implemented.
Wk 4 (22–28 Sep) — Full load. First OLAP queries running. Partitioning strategy tested.
Wk 5 (29 Sep–5 Oct) — Airflow DAG automated end to end. Quality metrics captured.
Wk 6 (6–12 Oct) — EDA, baseline model, query benchmarking, report written.

The binding constraint is 13 October. Geospatial ETL is front-loaded, so Workstream A starts on day one and Workstreams B and C should not wait for a complete pipeline before prototyping against a sample.

Open decisions we need to make

Region. Brazil (best ancillary data, MapBiomas cross-references available) versus Southeast Asia (closer to home, possibly more interesting to the audience). Affects everything downstream.
Classification labels. If we add a third mining task, where do labels come from? Options are the FIRMS `type` field, MapBiomas validated alerts, or a land-cover join. Vague labels are the most common way this kind of project stalls.
Warehouse engine. Postgres + PostGIS is the safe choice; DuckDB is faster to stand up and easier to containerise but weaker on spatial.

Known risks

Geospatial ETL is front-loaded. Every other team starts with a CSV that just loads. We start with coordinate reference systems, spatial indexing, and joins. If this slips two weeks, the October progress report has an empty warehouse. Budget for it explicitly.
Overlap with our GIS course. Reusing the domain and data is normally fine; reusing deliverables is not. Worth a short email to both instructors early.
WDPA historical snapshots depend on Earth Engine access. Confirm this works before the progress report, not after.

Immediate next steps

Pick the region — blocks everything else.
Pull the FIRMS country yearly summary and get the exact record count into the proposal.
Download one file from each source and verify the real column headers.
Draft the architecture diagram and division-of-work table.